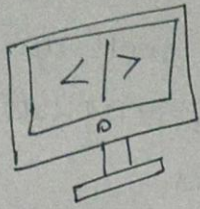


CD1 - ASSESSMENT TOOL 2

CONCEPT MAPPING

Q12: CONCEPT MAP- Algorithm $\rightarrow$ Best case, Average case, worst case $\rightarrow$ Time complexity $\rightarrow$ Performance evaluation

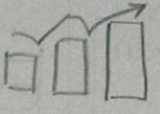


1. Algorithm
A finite set of instructions to solve a problem



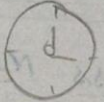
2. Best case

Best possible Input instance
Minimum running time

3. Average case

Average over all possible inputs
Expected running time

4. Worst case

Worst possible input instance
Maximum running time

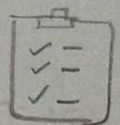
5. Time complexity

Describes how running time grows with input size n



6 Asymptotic Notation

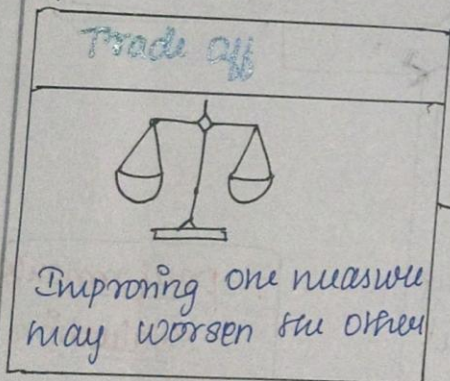
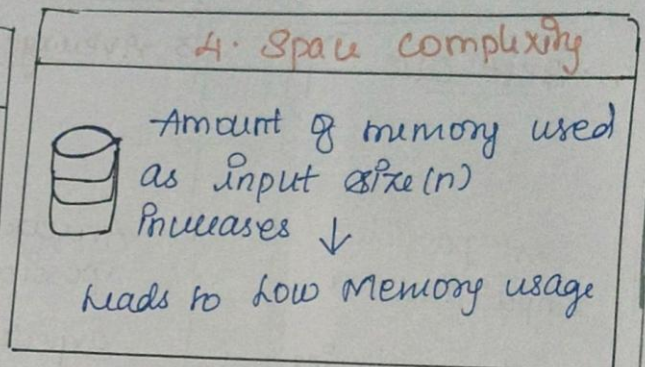
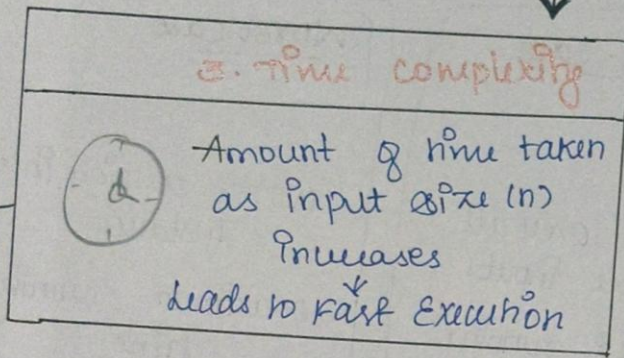
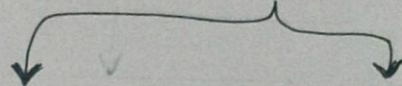
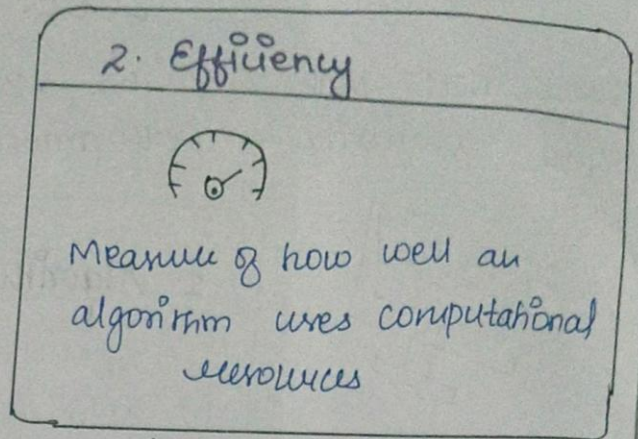
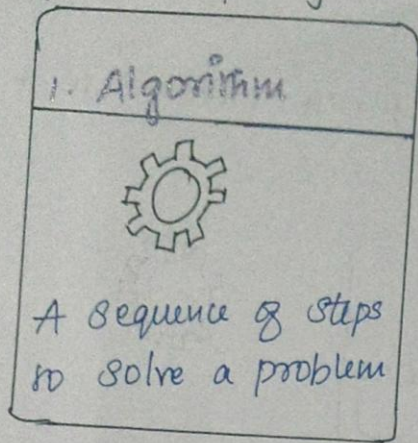
<u>Big O(O)</u>	<u>Big Ω(Omega)</u>	<u>Big Θ(Theta)</u>
<u>Upper Bound</u>	<u>Lower bound</u>	<u>Tight Bound</u>
$T(n) = O(g(n))$	$T(n) = \Omega(g(n))$	$T(n) = \Theta(g(n))$
Gives the maximum growth rate	Gives the minimum growth rate	Gives both upper & lower bounds

7. performance evaluation



used to compare algorithms & choose the most efficient one

Q28 CONCEPT MAP: Algorithm $\rightarrow$ Efficiency $\rightarrow$ Time complexity & Space complexity $\rightarrow$ Trade-off:



More memory $\rightarrow$ Faster execution
Extra memory can reduce computation time. Example: Caching.

Less memory $\rightarrow$ Slower execution
Limited memory may increase computation time. Example: In-place algorithms.

KEY TAKEAWAY



- Time complexity focuses on speed of execution.
- Space complexity focuses on memory usage.
- There is a trade-off between time & space.
- Efficient algorithm design balances both to meet real-world needs.

6. Algorithm Optimization & Design

Balance both factors to achieve optimal performance based on applications requirements